Steinitz exchange lemma

Lemma:

V : F-vector space

$Y, X \subseteq V$: finite subsets

s.t. (a) $\mathrm{Span}(X) = V$

(b) Y is linearly independent

Suppose that Y has m elements

& X has n elements

Then: (1) $m \le n$.

(2) If $Y = \{y_1, y_2, \ldots, y_m\}$.

then we have $X = \{x_1, x_2, \ldots, x_m, x_{m+1}, \ldots, x_n\}$.

s.t. the set $X' = \{\underbrace{y_1, y_2, \dots, y_m}, x_{m+1}, \dots, x_n\}$

also spans V (i.e. $\text{Span}(X') = V$)

Pf: Proof will be by induction on m

Base case ($m=0$) There is nothing to show

Inductive hypothesis Y' : lin. ind. subset $\cap \subseteq V$ with $m-1$

Then $Y' = \{y_1, y_2, \dots, y_{m-1}\}$

$$X = \{x_1, x_2, \dots, x_{m-1}, x_m, x_{m+1}, \dots, x_n\}$$

s.t. $X'' = \{y_1, y_2, \dots, y_{m-1}, x_m, \dots, x_n\}$

satisfies $\operatorname{Span}(X'') = V$

Inductive step:

$$Y = \{\underbrace{y_1, y_2, \dots, y_{m-1}}, \boxed{y_m}\}$$

$$Y' = \{y_1, y_2, \dots, y_{m-1}\}$$

$$X'' = \{y_1, y_2, \dots, y_{m-1}, \underbrace{x_m, \dots, x_n}\}$$

Need to know: We can exchange one of these elements with y_m without changing spanning property

We know:

(a) X'' spans V

i.e. Every $v \in V$ is a linear combination of

$y_1, \ldots, y_{m-1}, x_m, \ldots, x_n$

In particular

$$y_m = c_1 y_1 + c_2 y_2 + \cdots + c_{m-1} y_{m-1} + \boxed{c_m x_m + \cdots + c_n x_n}$$

for $c_1, \ldots, c_n \in F$

(b) Y is linearly independent

In particular,

y_m is <u>not</u> a linear combination of $y_1, y_2, \ldots, y_{m-1}$

$$y_m = c_1 y_1 + c_2 y_2 + \dots + c_{m-1} y_{m-1} + \boxed{c_m x_m + \dots + c_n x_n}$$

for $c_1, \dots, c_n \in F$

<u>and</u> at least one of $c_m, c_{m+1}, \dots, c_n$ is non-zero

Without loss of generality (WLOG) can assume $c_m \neq 0$

$\Big\Downarrow c_m^{-1}\cdot$

$$c_m^{-1} y_m = c_m^{-1} c_1 y_1 + c_m^{-1} c_2 y_2 + \dots + c_m^{-1} c_{m-1} y_{m-1} + x_m + \dots + c_m^{-1} c_n x_n$$

$$\Downarrow$$

$$x_m = -c_m^{-1} c_1 y_1 - c_m^{-1} c_2 y_2 - \dots - c_m^{-1} c_{m-1} y_{m-1} + c_m^{-1} y_m - c_m^{-1} c_{m+1} x_{m+1} - \dots - c_m^{-1} c_n x_n$$

$\Rightarrow$ x_m is a linear combination of $\underbrace{y_1, \dots, y_m, x_{m+1}, \dots, x_n}$

This tells me

$$\operatorname{Span}\Big(\overbrace{\{y_1, \dots, y_{m-1}, x_m, x_{m+1}, \dots, x_n\}}^{X''}\Big)$$

$$\subseteq \operatorname{span}\big(\{y_1, \dots, y_{m-1}, y_m, x_{m+1}, \dots, x_n\}\big)$$

$$\Rightarrow \quad V = \mathrm{Span}(X'')$$
$$\subseteq \mathrm{Span}(X')$$
$$\subseteq V$$

$$\Rightarrow \quad \mathrm{Span}(X') = V$$

Q.E.D.

$\blacksquare$

Defn V is a finite dimensional F-vector space if the following equivalent conditions hold:

(a) $\exists$ a <u>finite</u> subset $X \subseteq V$ s.t.

$$\mathrm{Span}(X) = V$$

(b) V admits a finite basis

Rmk (1) Equivalence follows from Corollary (1) from previous lecture

Rmk ② In this case, every basis for V has the same number of elements

Def^n V: $\underbrace{\text{finite dimensional}}_{\text{f.d.}}$

The **dimension of V**, denoted **$\dim(V)$**, is the size of any basis for V.

Example:

1. $\dim(F^n) = n$
2. $\dim(\mathrm{Poly}_{\leq m}(F)) = m+1$

A basis is given by

$$\underbrace{\{1, x, x^2, \ldots, x^m\}}_{m+1 \text{ elements}}$$

3. $\dim(M_{r \times s}(F)) = r \cdot s$

A basis is given by

$$\left\{ \underbrace{E_{i,j}} : \begin{array}{l} 1 \leq i \leq r \\ 1 \leq j \leq s \end{array} \right\}$$

The matrix with 1 in the i-th row

& j-th
column

and 0s everywhere else

$r=3,\ s=2$

$$E_{1,2} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Observation ①

V : f.d. F-vector space

$W \subseteq V$: subspace

W is also finite dimensional

& $\dim(W) \leq \dim(V)$

Observation ②

In fact, if $\{w_1, w_2, \ldots, w_m\}$ is a basis for W then, we can **extend** this to a basis

$\{w_1, w_2, \ldots, w_m, v_{m+1}, \ldots, v_n\}$

for V

Pf: Consequence of the exchange lemma

Observation (3)

$\dim(V) = n$

$\{v_1, v_2, \ldots, v_n\} \subseteq V$

TFAE: (1) $\{v_1, v_2, \ldots, v_n\}$ is a basis

(2) $\{v_1, v_2, \ldots, v_n\}$ is linearly independent

(3) $\{v_1, v_2, \ldots, v_n\}$ spans V

Pf: WTS: $(2) \Longleftrightarrow (3)$

"is equivalent to"

We know: There exists a basis

$\{w_1, w_2, .., w_n\}$ for V

(This is the meaning of $\dim(V) = n$)

$\boxed{(2) \Rightarrow (3)}$

If $\{v_1, v_2, .., v_n\}$ is linearly independent then the Steinitz Exchange Lemma applied with $Y = \{v_1, v_2, .. v_n\}$

$X = \{w_1, w_2, -, w_n\}$

tells us $X' = \{v_1, v_2, .., v_n\}$ (all w_i's exchanged!!)

$= Y$ (!!)

satisfies $\mathrm{span}(Y) = \mathrm{span}(X') = \mathrm{span}(X) = V$

$\boxed{(3) \Rightarrow (2)}$ If $\mathrm{span}(\{v_1, v_2, .., v_n\}) = V$

we claim $\{v_1, v_2, \ldots, v_n\}$ is linearly independent

Otherwise, WLOG, v_n is a linear combination of $v_1, v_2, \ldots, v_{n-1}$

But then

$$\mathrm{Span}(\{v_1, v_2, \ldots, v_{n-1}\}) = \mathrm{Span}(\{v_1, v_2, \ldots, v_n\})$$

(why is this?)

This now tells us that V can be spanned by the $n-1$ elements

$v_1, v_2, \ldots, v_{n-1}$

But now applying Steinitz Exchange Lemma to

$X = \{v_1, v_2, \ldots, v_{n-1}\}$

$Y = \{w_1, w_2, \ldots, w_n\}$

we get the absurdity

$n \leq n-1$ XX

IMPOSSIBLE!

Def^{n} V: f.d. F-vector space

$\mathcal{B}_1 = \{v_1, v_2, \dots, v_n\}$

$\mathcal{B}_2 = \{w_1, w_2, \dots, w_n\}$

are two bases for V

The change of basis matrix from $\mathcal{B}_1$ to $\mathcal{B}_2$ is the matrix $A \in M_{n\times n}(F)$ determined as follows:

$$A = (a_{i,j})_{\substack{1\le i\le n \\ 1\le j\le n}}$$

where $a_{i,j}$ are determined by.

$$W_i = a_{i,1}V_1 + a_{i,2}V_2 + \dots + a_{i,n}V_n$$

$1 \le i \le n$